

BANK DATABASE - SQL VIEWS

1. CREATE / SELECT DATABASE

```
[mysql> USE BankDB;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
[mysql> SHOW TABLES;
+-----+
| Tables_in_bankdb |
+-----+
| Account           |
| Bank_Transaction  |
| Customer          |
| Loan              |
| Transaction_Audit |
+-----+
5 rows in set (0.001 sec)
```

2. CHECK EXISTING CUSTOMER DATA

```
[mysql> SELECT * FROM Customer;
+-----+-----+-----+-----+-----+
| Customer_ID | Customer_Name | Phone   | Email           | City       |
+-----+-----+-----+-----+-----+
| 101         | Ravi Kumar    | 9876543210 | ravi@gmail.com  | Hyderabad |
| 102         | Priya Sharma  | 9876543211 | priya@gmail.com | Vijayawada |
| 103         | Arjun Reddy   | 9876543212 | arjun@gmail.com | Bangalore  |
| 104         | Sneha Rao     | 9876543213 | sneha@gmail.com | Chennai    |
| 105         | Kiran Kumar   | 9876543214 | kiran@gmail.com | Hyderabad  |
| 106         | Anil Kumar    | 9876543215 | anil@gmail.com  | Delhi      |
| 107         | Meena Reddy   | 9876543216 | meena@gmail.com | Mumbai     |
| 108         | Rahul Sharma  | 9876543217 | rahul@gmail.com | Pune       |
| 109         | Lakshmi Devi  | 9876543218 | lakshmi@gmail.com | Hyderabad  |
| 110         | Suresh Babu   | 9876543219 | suresh@gmail.com | Vijayawada |
+-----+-----+-----+-----+-----+
10 rows in set (0.000 sec)
```

3. ADD THE ADDITIONAL ACCOUNT DATA

```
[mysql> SELECT * FROM Account;
+-----+-----+-----+-----+-----+
| Account_No | Customer_ID | Account_Type | Balance   | Branch     |
+-----+-----+-----+-----+-----+
| 10001      | 101         | Savings      | 52000.00 | Hyderabad  |
| 10002      | 102         | Savings      | 80000.00 | Vijayawada |
| 10003      | 103         | Current      | 120000.00 | Bangalore  |
| 10004      | 104         | Savings      | 45000.00 | Chennai    |
| 10005      | 105         | Current      | 90000.00 | Hyderabad  |
| 10006      | 106         | Savings      | 65000.00 | Delhi      |
| 10007      | 107         | Current      | 150000.00 | Mumbai     |
| 10008      | 108         | Savings      | 35000.00 | Pune       |
| 10009      | 109         | Savings      | 85000.00 | Hyderabad  |
| 10010      | 110         | Current      | 110000.00 | Vijayawada |
+-----+-----+-----+-----+-----+
10 rows in set (0.000 sec)
```

4. ADD THE ADDITIONAL TRANSACTION DATA

```
[mysql> SELECT * FROM Bank_Transaction;
```

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-18 14:04:27
2	10002	DEPOSIT	15000.00	2026-08-18 14:04:27
3	10003	WITHDRAW	20000.00	2026-08-18 14:04:27
4	10004	DEPOSIT	5000.00	2026-08-18 14:04:27
5	10005	WITHDRAW	10000.00	2026-08-18 14:04:27
6	10001	DEPOSIT	2500.00	2026-08-18 14:38:13
7	10001	DEPOSIT	5000.00	2026-08-18 14:41:01
8	10006	DEPOSIT	12000.00	2026-08-18 17:36:24
9	10007	DEPOSIT	25000.00	2026-08-18 17:36:24
10	10008	WITHDRAW	5000.00	2026-08-18 17:36:24
11	10009	DEPOSIT	20000.00	2026-08-18 17:36:24
12	10010	WITHDRAW	15000.00	2026-08-18 17:36:24

```
12 rows in set (0.000 sec)
```

5. ADD THE ADDITIONAL LOAN DATA

```
[mysql> SELECT * FROM Loan;
```

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50
505	105	Home Loan	4000000.00	7.20
506	106	Car Loan	900000.00	8.50
507	107	Business Loan	3000000.00	9.00
508	109	Personal Loan	600000.00	10.00

```
8 rows in set (0.000 sec)
```

6. DISPLAY ORIGINAL TABLES

7) Customer Table

```
[mysql> SELECT * FROM Customer;
```

Customer_ID	Customer_Name	Phone	Email	City
101	Ravi Kumar	9876543210	ravi@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad
106	Anil Kumar	9876543215	anil@gmail.com	Delhi
107	Meena Reddy	9876543216	meena@gmail.com	Mumbai
108	Rahul Sharma	9876543217	rahul@gmail.com	Pune
109	Lakshmi Devi	9876543218	lakshmi@gmail.com	Hyderabad
110	Suresh Babu	9876543219	suresh@gmail.com	Vijayawada

```
10 rows in set (0.000 sec)
```

8) Account Table

```
mysql> SELECT * FROM Account;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	52000.00	Hyderabad
10002	102	Savings	80000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
10006	106	Savings	77000.00	Delhi
10007	107	Current	175000.00	Mumbai
10008	108	Savings	30000.00	Pune
10009	109	Savings	105000.00	Hyderabad
10010	110	Current	95000.00	Vijayawada

10 rows in set (0.000 sec)

9) Bank Transaction Table

```
mysql> SELECT * FROM Bank_Transaction;
```

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-18 14:04:27
2	10002	DEPOSIT	15000.00	2026-08-18 14:04:27
3	10003	WITHDRAW	20000.00	2026-08-18 14:04:27
4	10004	DEPOSIT	5000.00	2026-08-18 14:04:27
5	10005	WITHDRAW	10000.00	2026-08-18 14:04:27
6	10001	DEPOSIT	2500.00	2026-08-18 14:38:13
7	10001	DEPOSIT	5000.00	2026-08-18 14:41:01
8	10006	DEPOSIT	12000.00	2026-08-18 17:36:24
9	10007	DEPOSIT	25000.00	2026-08-18 17:36:24
10	10008	WITHDRAW	5000.00	2026-08-18 17:36:24
11	10009	DEPOSIT	20000.00	2026-08-18 17:36:24
12	10010	WITHDRAW	15000.00	2026-08-18 17:36:24

12 rows in set (0.000 sec)

10) Loan Table

```
mysql> SELECT * FROM Loan;
```

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50
505	105	Home Loan	4000000.00	7.20
506	106	Car Loan	900000.00	8.50
507	107	Business Loan	3000000.00	9.00
508	109	Personal Loan	600000.00	10.00

8 rows in set (0.000 sec)

PART A - SIMPLE VIEWS

11. CREATE A VIEW FOR ALL CUSTOMERS

```
mysql> CREATE VIEW Customer_View AS
-> SELECT *
-> FROM Customer;
Query OK, 0 rows affected (0.012 sec)

mysql> SELECT * FROM Customer_View;
```

Customer_ID	Customer_Name	Phone	Email	City
101	Ravi Kumar	9876543210	ravi@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad
106	Anil Kumar	9876543215	anil@gmail.com	Delhi
107	Meena Reddy	9876543216	meena@gmail.com	Mumbai
108	Rahul Sharma	9876543217	rahul@gmail.com	Pune
109	Lakshmi Devi	9876543218	lakshmi@gmail.com	Hyderabad
110	Suresh Babu	9876543219	suresh@gmail.com	Vijayawada

```
10 rows in set (0.002 sec)
```

12. CREATE A VIEW FOR CUSTOMER BASIC INFORMATION

```
mysql> CREATE VIEW Customer_Basic_View AS
-> SELECT
->     Customer_ID,
->     Customer_Name,
->     City
-> FROM Customer;
Query OK, 0 rows affected (0.003 sec)

mysql> SELECT * FROM Customer_Basic_View;
```

Customer_ID	Customer_Name	City
101	Ravi Kumar	Hyderabad
102	Priya Sharma	Vijayawada
103	Arjun Reddy	Bangalore
104	Sneha Rao	Chennai
105	Kiran Kumar	Hyderabad
106	Anil Kumar	Delhi
107	Meena Reddy	Mumbai
108	Rahul Sharma	Pune
109	Lakshmi Devi	Hyderabad
110	Suresh Babu	Vijayawada

```
10 rows in set (0.001 sec)
```

13. CREATE A VIEW FOR ACCOUNT INFORMATION

```
mysql> CREATE VIEW Account_View AS
-> SELECT
->     Account_No,
->     Account_Type,
->     Balance,
->     Branch
[ -> FROM Account;
Query OK, 0 rows affected (0.004 sec)
```

```
[mysql> SELECT * FROM Account_View;
```

Account_No	Account_Type	Balance	Branch
10001	Savings	52000.00	Hyderabad
10002	Savings	80000.00	Vijayawada
10003	Current	120000.00	Bangalore
10004	Savings	45000.00	Chennai
10005	Current	90000.00	Hyderabad
10006	Savings	77000.00	Delhi
10007	Current	175000.00	Mumbai
10008	Savings	30000.00	Pune
10009	Savings	105000.00	Hyderabad
10010	Current	95000.00	Vijayawada

10 rows in set (0.001 sec)

14. CREATE A VIEW FOR SAVINGS ACCOUNTS

```
[mysql> SELECT * FROM Savings_Account_View;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	52000.00	Hyderabad
10002	102	Savings	80000.00	Vijayawada
10004	104	Savings	45000.00	Chennai
10006	106	Savings	77000.00	Delhi
10008	108	Savings	30000.00	Pune
10009	109	Savings	105000.00	Hyderabad

6 rows in set (0.003 sec)

15. CREATE A VIEW FOR CURRENT ACCOUNTS

```
[mysql> SELECT * FROM Current_Account_View;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10003	103	Current	120000.00	Bangalore
10005	105	Current	90000.00	Hyderabad
10007	107	Current	175000.00	Mumbai
10010	110	Current	95000.00	Vijayawada

4 rows in set (0.001 sec)

PART B — VIEWS WITH CONDITIONS

16. HIGH BALANCE ACCOUNT VIEW

```
[mysql> SELECT * FROM High_Balance_View;
+-----+-----+-----+-----+
| Account_No | Customer_ID | Account_Type | Balance |
+-----+-----+-----+-----+
|      10003 |          103 | Current      | 120000.00 |
|      10007 |          107 | Current      | 175000.00 |
|      10009 |          109 | Savings      | 105000.00 |
+-----+-----+-----+-----+
3 rows in set (0.002 sec)
```

17. HYDERABAD ACCOUNT VIEW

```
[mysql> SELECT * FROM Hyderabad_Account_View;
+-----+-----+-----+-----+-----+
| Account_No | Customer_ID | Account_Type | Balance | Branch |
+-----+-----+-----+-----+-----+
|      10001 |          101 | Savings      | 52000.00 | Hyderabad |
|      10005 |          105 | Current      | 90000.00 | Hyderabad |
|      10009 |          109 | Savings      | 105000.00 | Hyderabad |
+-----+-----+-----+-----+-----+
3 rows in set (0.001 sec)
```

18. LOW BALANCE ACCOUNT VIEW

```
mysql> CREATE VIEW Low_Balance_View AS
-> SELECT
->     Account_No,
->     Customer_ID,
->     Balance
-> FROM Account
-> WHERE Balance < 50000;
Query OK, 0 rows affected (0.003 sec)

[mysql> SELECT * FROM Low_Balance_View;
+-----+-----+-----+
| Account_No | Customer_ID | Balance |
+-----+-----+-----+
|      10004 |          104 | 45000.00 |
|      10008 |          108 | 30000.00 |
+-----+-----+-----+
2 rows in set (0.002 sec)
```


PART C — VIEWS USING JOINS

19. CUSTOMER + ACCOUNT VIEW

```
mysql> SELECT * FROM Customer_Account_View;
```

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch
101	Ravi Kumar	Hyderabad	10001	Savings	52000.00	Hyderabad
102	Priya Sharma	Vijayawada	10002	Savings	80000.00	Vijayawada
103	Arjun Reddy	Bangalore	10003	Current	120000.00	Bangalore
104	Sneha Rao	Chennai	10004	Savings	45000.00	Chennai
105	Kiran Kumar	Hyderabad	10005	Current	90000.00	Hyderabad
106	Anil Kumar	Delhi	10006	Savings	77000.00	Delhi
107	Meena Reddy	Mumbai	10007	Current	175000.00	Mumbai
108	Rahul Sharma	Pune	10008	Savings	30000.00	Pune
109	Lakshmi Devi	Hyderabad	10009	Savings	105000.00	Hyderabad
110	Suresh Babu	Vijayawada	10010	Current	95000.00	Vijayawada

10 rows in set (0.001 sec)

20. HYDERABAD CUSTOMER ACCOUNT VIEW

```
mysql> SELECT * FROM Hyderabad_Customer_Accounts;
```

Customer_Name	City	Account_No	Account_Type	Balance
Ravi Kumar	Hyderabad	10001	Savings	52000.00
Kiran Kumar	Hyderabad	10005	Current	90000.00
Lakshmi Devi	Hyderabad	10009	Savings	105000.00

3 rows in set (0.003 sec)

21. CUSTOMER LOAN VIEW

```
mysql> SELECT * FROM Customer_Loan_View;
```

Customer_ID	Customer_Name	City	Loan_ID	Loan_Type	Loan_Amount	Interest_Rate
101	Ravi Kumar	Hyderabad	501	Home Loan	5000000.00	7.50
102	Priya Sharma	Vijayawada	502	Education Loan	1000000.00	6.50
103	Arjun Reddy	Bangalore	503	Car Loan	800000.00	8.20
104	Sneha Rao	Chennai	504	Personal Loan	500000.00	10.50
105	Kiran Kumar	Hyderabad	505	Home Loan	4000000.00	7.20
106	Anil Kumar	Delhi	506	Car Loan	900000.00	8.50
107	Meena Reddy	Mumbai	507	Business Loan	3000000.00	9.00
109	Lakshmi Devi	Hyderabad	508	Personal Loan	600000.00	10.00

8 rows in set (0.001 sec)

22. COMPLETE CUSTOMER BANKING VIEW

```
mysql> SELECT * FROM Customer_Banking_View;
```

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch	Loan_Type	Loan_Amount
101	Ravi Kumar	Hyderabad	10001	Savings	52000.00	Hyderabad	Home Loan	5000000.00
102	Priya Sharma	Vijayawada	10002	Savings	80000.00	Vijayawada	Education Loan	1000000.00
103	Arjun Reddy	Bangalore	10003	Current	120000.00	Bangalore	Car Loan	800000.00
104	Sneha Rao	Chennai	10004	Savings	45000.00	Chennai	Personal Loan	500000.00
105	Kiran Kumar	Hyderabad	10005	Current	90000.00	Hyderabad	Home Loan	4000000.00
106	Anil Kumar	Delhi	10006	Savings	77000.00	Delhi	Car Loan	900000.00
107	Meena Reddy	Mumbai	10007	Current	175000.00	Mumbai	Business Loan	3000000.00
108	Rahul Sharma	Pune	10008	Savings	30000.00	Pune	NULL	NULL
109	Lakshmi Devi	Hyderabad	10009	Savings	105000.00	Hyderabad	Personal Loan	600000.00
110	Suresh Babu	Vijayawada	10010	Current	95000.00	Vijayawada	NULL	NULL

10 rows in set (0.001 sec)

PART D — VIEWS USING AGGREGATE FUNCTIONS

23. TOTAL BANK BALANCE

```
[mysql> SELECT * FROM Total_Bank_Balance;
+-----+
| Total_Balance |
+-----+
|      869000.00 |
+-----+
1 row in set (0.003 sec)
```

24. AVERAGE ACCOUNT BALANCE

```
[mysql> SELECT * FROM Average_Account_Balance;
+-----+
| Average_Balance |
+-----+
|      86900.000000 |
+-----+
1 row in set (0.001 sec)
```

25. MAXIMUM ACCOUNT BALANCE

```
[mysql> SELECT * FROM Maximum_Balance_View;
+-----+
| Maximum_Balance |
+-----+
|      175000.00 |
+-----+
1 row in set (0.001 sec)
```

26. MINIMUM ACCOUNT BALANCE

```
[mysql> SELECT * FROM Minimum_Balance_View;
+-----+
| Minimum_Balance |
+-----+
|       30000.00 |
+-----+
1 row in set (0.001 sec)
```


27. TOTAL NUMBER OF ACCOUNTS

```
[mysql> SELECT * FROM Account_Count_View;
+-----+
| Total_Accounts |
+-----+
|          10    |
+-----+
1 row in set (0.001 sec)
```

PART E – GROUP BY VIEWS

28. BRANCH-WISE ACCOUNT COUNT

```
[mysql> SELECT * FROM Branch_Account_Count;
+-----+-----+
| Branch      | Number_of_Accounts |
+-----+-----+
| Hyderabad   |          3          |
| Vijayawada   |          2          |
| Bangalore    |          1          |
| Chennai      |          1          |
| Delhi        |          1          |
| Mumbai       |          1          |
| Pune         |          1          |
+-----+-----+
7 rows in set (0.003 sec)
```

29. BRANCH-WISE TOTAL BALANCE

```
[mysql> SELECT * FROM Branch_Total_Balance;
+-----+-----+
| Branch      | Total_Balance |
+-----+-----+
| Hyderabad   | 247000.00     |
| Vijayawada   | 175000.00     |
| Bangalore    | 120000.00     |
| Chennai      | 45000.00      |
| Delhi        | 77000.00      |
| Mumbai       | 175000.00     |
| Pune         | 30000.00      |
+-----+-----+
7 rows in set (0.001 sec)
```

30. ACCOUNT TYPE COUNT

```
[mysql> SELECT * FROM Account_Type_Count;
+-----+-----+
| Account_Type | Number_of_Accounts |
+-----+-----+
| Savings      | 6                   |
| Current      | 4                   |
+-----+-----+
2 rows in set (0.001 sec)
```

31. ACCOUNT TYPE TOTAL BALANCE

```
[mysql> SELECT * FROM Account_Type_Balance;
+-----+-----+
| Account_Type | Total_Balance |
+-----+-----+
| Savings      | 389000.00     |
| Current      | 480000.00     |
+-----+-----+
2 rows in set (0.002 sec)
```

PART F — HAVING IN VIEWS

32. BRANCHES WITH MORE THAN ONE ACCOUNT

```
[mysql> SELECT * FROM Multiple_Account_Branches;
+-----+-----+
| Branch      | Number_of_Accounts |
+-----+-----+
| Hyderabad   | 3                   |
| Vijayawada   | 2                   |
+-----+-----+
2 rows in set (0.001 sec)
```

33. BRANCHES WITH TOTAL BALANCE ABOVE 100000

```
[mysql> SELECT * FROM Rich_Branches;
+-----+-----+
| Branch      | Total_Balance |
+-----+-----+
| Hyderabad   | 247000.00     |
| Vijayawada   | 175000.00     |
| Bangalore    | 120000.00     |
| Mumbai      | 175000.00     |
+-----+-----+
4 rows in set (0.002 sec)
```

PART G — TRANSACTION VIEWS

34. DEPOSIT TRANSACTION VIEW

```
mysql> SELECT * FROM Deposit_Transaction_View;
```

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-18 14:04:27
2	10002	DEPOSIT	15000.00	2026-08-18 14:04:27
4	10004	DEPOSIT	5000.00	2026-08-18 14:04:27
6	10001	DEPOSIT	2500.00	2026-08-18 14:38:13
7	10001	DEPOSIT	5000.00	2026-08-18 14:41:01
8	10006	DEPOSIT	12000.00	2026-08-18 17:36:24
9	10007	DEPOSIT	25000.00	2026-08-18 17:36:24
11	10009	DEPOSIT	20000.00	2026-08-18 17:36:24

8 rows in set (0.001 sec)

35. WITHDRAWAL TRANSACTION VIEW

```
mysql> SELECT * FROM Withdrawal_Transaction_View;
```

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
3	10003	WITHDRAW	20000.00	2026-08-18 14:04:27
5	10005	WITHDRAW	10000.00	2026-08-18 14:04:27
10	10008	WITHDRAW	5000.00	2026-08-18 17:36:24
12	10010	WITHDRAW	15000.00	2026-08-18 17:36:24

4 rows in set (0.001 sec)

36. TRANSACTION DETAILS WITH CUSTOMER

```
mysql> SELECT * FROM Customer_Transaction_View;
```

Customer_Name	Account_No	Account_Type	Transaction_ID	Transaction_Type	Amount	Transaction_Date
Ravi Kumar	10001	Savings	1	DEPOSIT	10000.00	2026-08-18 14:04:27
Ravi Kumar	10001	Savings	6	DEPOSIT	2500.00	2026-08-18 14:38:13
Ravi Kumar	10001	Savings	7	DEPOSIT	5000.00	2026-08-18 14:41:01
Priya Sharma	10002	Savings	2	DEPOSIT	15000.00	2026-08-18 14:04:27
Arjun Reddy	10003	Current	3	WITHDRAW	20000.00	2026-08-18 14:04:27
Sneha Rao	10004	Savings	4	DEPOSIT	5000.00	2026-08-18 14:04:27
Kiran Kumar	10005	Current	5	WITHDRAW	10000.00	2026-08-18 14:04:27
Anil Kumar	10006	Savings	8	DEPOSIT	12000.00	2026-08-18 17:36:24
Meena Reddy	10007	Current	9	DEPOSIT	25000.00	2026-08-18 17:36:24
Rahul Sharma	10008	Savings	10	WITHDRAW	5000.00	2026-08-18 17:36:24
Lakshmi Devi	10009	Savings	11	DEPOSIT	20000.00	2026-08-18 17:36:24
Suresh Babu	10010	Current	12	WITHDRAW	15000.00	2026-08-18 17:36:24

12 rows in set (0.002 sec)

37. HIGH VALUE TRANSACTION VIEW

```
mysql> SELECT * FROM High_Value_Transaction_View;
```

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
2	10002	DEPOSIT	15000.00	2026-08-18 14:04:27
3	10003	WITHDRAW	20000.00	2026-08-18 14:04:27
8	10006	DEPOSIT	12000.00	2026-08-18 17:36:24
9	10007	DEPOSIT	25000.00	2026-08-18 17:36:24
11	10009	DEPOSIT	20000.00	2026-08-18 17:36:24
12	10010	WITHDRAW	15000.00	2026-08-18 17:36:24

6 rows in set (0.001 sec)

PART H – VIEWS WITH ORDER BY

38. ACCOUNTS ORDERED BY BALANCE

```
[mysql> SELECT * FROM Balance_Ranking_View;
```

Account_No	Customer_ID	Account_Type	Balance
10007	107	Current	175000.00
10003	103	Current	120000.00
10009	109	Savings	105000.00
10010	110	Current	95000.00
10005	105	Current	90000.00
10002	102	Savings	80000.00
10006	106	Savings	77000.00
10001	101	Savings	52000.00
10004	104	Savings	45000.00
10008	108	Savings	30000.00

```
10 rows in set (0.002 sec)
```

39. CUSTOMERS ORDERED BY NAME

```
[mysql> SELECT * FROM Customer_Name_View;
```

Customer_ID	Customer_Name	City
106	Anil Kumar	Delhi
103	Arjun Reddy	Bangalore
105	Kiran Kumar	Hyderabad
109	Lakshmi Devi	Hyderabad
107	Meena Reddy	Mumbai
102	Priya Sharma	Vijayawada
108	Rahul Sharma	Pune
101	Ravi Kumar	Hyderabad
104	Sneha Rao	Chennai
110	Suresh Babu	Vijayawada

```
10 rows in set (0.002 sec)
```

PART I — VIEW WITH CALCULATED COLUMN

40. LOAN INTEREST CALCULATION VIEW

```
mysql> SELECT * FROM Loan_Interest_View;
```

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate	Annual_Interest
501	101	Home Loan	500000.00	7.50	375000.00000000
502	102	Education Loan	100000.00	6.50	65000.00000000
503	103	Car Loan	80000.00	8.20	65600.00000000
504	104	Personal Loan	50000.00	10.50	52500.00000000
505	105	Home Loan	400000.00	7.20	288000.00000000
506	106	Car Loan	90000.00	8.50	76500.00000000
507	107	Business Loan	300000.00	9.00	270000.00000000
508	109	Personal Loan	60000.00	10.00	60000.00000000

8 rows in set (0.001 sec)

41. LOAN TOTAL AMOUNT VIEW

```
mysql> SELECT * FROM Loan_Total_Amount_View;
```

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate	Total_Amount
501	101	Home Loan	500000.00	7.50	5375000.00000000
502	102	Education Loan	100000.00	6.50	1065000.00000000
503	103	Car Loan	80000.00	8.20	865600.00000000
504	104	Personal Loan	50000.00	10.50	552500.00000000
505	105	Home Loan	400000.00	7.20	4288000.00000000
506	106	Car Loan	90000.00	8.50	976500.00000000
507	107	Business Loan	300000.00	9.00	3270000.00000000
508	109	Personal Loan	60000.00	10.00	660000.00000000

8 rows in set (0.001 sec)

PART J — USING VIEW WITH QUERIES

42. DISPLAY HIGH BALANCE ACCOUNTS FROM VIEW

```
mysql> SELECT *  
-> FROM High_Balance_View  
-> WHERE Balance > 120000;
```

Account_No	Customer_ID	Account_Type	Balance
10007	107	Current	175000.00

1 row in set (0.001 sec)

43. DISPLAY SAVINGS ACCOUNTS ABOVE 60000

```
mysql> SELECT *
-> FROM Savings_Account_View
-> WHERE Balance > 60000;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10002	102	Savings	80000.00	Vijayawada
10006	106	Savings	77000.00	Delhi
10009	109	Savings	105000.00	Hyderabad

3 rows in set (0.001 sec)

44. DISPLAY HYDERABAD ACCOUNTS ABOVE 50000

```
mysql> SELECT *
-> FROM Hyderabad_Account_View
-> WHERE Balance > 50000;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	52000.00	Hyderabad
10005	105	Current	90000.00	Hyderabad
10009	109	Savings	105000.00	Hyderabad

3 rows in set (0.001 sec)

45. DISPLAY CUSTOMERS FROM HYDERABAD

```
mysql> SELECT *
-> FROM Customer_Basic_View
-> WHERE City = 'Hyderabad';
```

Customer_ID	Customer_Name	City
101	Ravi Kumar	Hyderabad
105	Kiran Kumar	Hyderabad
109	Lakshmi Devi	Hyderabad

3 rows in set (0.002 sec)

46. DISPLAY CUSTOMER ACCOUNTS ORDERED BY BALANCE

```
mysql> SELECT *
      -> FROM Customer_Account_View
      -> ORDER BY Balance DESC;
```

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance	Branch
107	Meena Reddy	Mumbai	10007	Current	175000.00	Mumbai
103	Arjun Reddy	Bangalore	10003	Current	120000.00	Bangalore
109	Lakshmi Devi	Hyderabad	10009	Savings	105000.00	Hyderabad
110	Suresh Babu	Vijayawada	10010	Current	95000.00	Vijayawada
105	Kiran Kumar	Hyderabad	10005	Current	90000.00	Hyderabad
102	Priya Sharma	Vijayawada	10002	Savings	80000.00	Vijayawada
106	Anil Kumar	Delhi	10006	Savings	77000.00	Delhi
101	Ravi Kumar	Hyderabad	10001	Savings	52000.00	Hyderabad
104	Sneha Rao	Chennai	10004	Savings	45000.00	Chennai
108	Rahul Sharma	Pune	10008	Savings	30000.00	Pune

10 rows in set (0.001 sec)

47. DISPLAY LOANS ABOVE 1000000

```
mysql> SELECT *
      -> FROM Customer_Loan_View
      -> WHERE Loan_Amount > 1000000;
```

Customer_ID	Customer_Name	City	Loan_ID	Loan_Type	Loan_Amount	Interest_Rate
101	Ravi Kumar	Hyderabad	501	Home Loan	5000000.00	7.50
105	Kiran Kumar	Hyderabad	505	Home Loan	4000000.00	7.20
107	Meena Reddy	Mumbai	507	Business Loan	3000000.00	9.00

3 rows in set (0.003 sec)

PART K — UPDATING A VIEW

48. UPDATE ACCOUNT BALANCE THROUGH VIEW

```
mysql> UPDATE Account_View
      -> SET Balance = 60000
      -> WHERE Account_No = 10001;
Query OK, 1 row affected (0.008 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT *
      -> FROM Account
      -> WHERE Account_No = 10001;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	60000.00	Hyderabad

1 row in set (0.000 sec)

PART L — INSERT THROUGH VIEW

49. INSERT USING SIMPLE VIEW

```
mysql> INSERT INTO Simple_Account_View  
-> VALUES  
[ -> (10011, 101, 'Savings', 55000, 'Hyderabad');  
Query OK, 1 row affected (0.001 sec)
```

```
mysql> SELECT *  
[ -> FROM Account;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	60000.00	Hyderabad
10002	102	Savings	80000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
10006	106	Savings	77000.00	Delhi
10007	107	Current	175000.00	Mumbai
10008	108	Savings	30000.00	Pune
10009	109	Savings	105000.00	Hyderabad
10010	110	Current	95000.00	Vijayawada
10011	101	Savings	55000.00	Hyderabad

11 rows in set (0.000 sec)

PART M — DELETE THROUGH VIEW

50. DELETE USING SIMPLE VIEW

```
mysql> DELETE FROM Simple_Account_View  
[ -> WHERE Account_No = 10011;  
Query OK, 1 row affected (0.005 sec)
```

```
[mysql> SELECT * FROM Account;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	60000.00	Hyderabad
10002	102	Savings	80000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
10006	106	Savings	77000.00	Delhi
10007	107	Current	175000.00	Mumbai
10008	108	Savings	30000.00	Pune
10009	109	Savings	105000.00	Hyderabad
10010	110	Current	95000.00	Vijayawada

10 rows in set (0.001 sec)

PART N — VIEW INFORMATION

51. SHOW ALL VIEWS

```
mysql> SHOW FULL TABLES
      -> WHERE TABLE_TYPE = 'VIEW';
```

Tables_in_bankdb	Table_type
account_count_view	VIEW
account_type_balance	VIEW
account_type_count	VIEW
account_view	VIEW
average_account_balance	VIEW
balance_ranking_view	VIEW
branch_account_count	VIEW
branch_total_balance	VIEW
current_account_view	VIEW
customer_account_view	VIEW
customer_banking_view	VIEW
customer_basic_view	VIEW
customer_loan_view	VIEW
customer_name_view	VIEW
customer_transaction_view	VIEW
customer_view	VIEW
deposit_transaction_view	VIEW
high_balance_view	VIEW
high_value_transaction_view	VIEW
hyderabad_account_view	VIEW
hyderabad_customer_accounts	VIEW
loan_interest_view	VIEW
loan_total_amount_view	VIEW
low_balance_view	VIEW
maximum_balance_view	VIEW
minimum_balance_view	VIEW
multiple_account_branches	VIEW
rich_branches	VIEW
savings_account_view	VIEW
simple_account_view	VIEW
total_bank_balance	VIEW
withdrawal_transaction_view	VIEW

32 rows in set (0.009 sec)

52. SHOW VIEW DEFINITION

```
mysql> SHOW CREATE VIEW Customer_Account_View;
+-----+-----+
| View | Create View |
+-----+-----+
|      | character_set_client | collation_connection |
+-----+-----+
| customer_account_view | CREATE ALGORITHM=UNDEFINED DEFINER='root'@'localhost' SQL SECURITY DEFINER VIEW 'customer_account_view' AS select 'C'.'Customer_ID' AS 'Customer_ID','C'.'Customer_Name' AS 'Customer_Name','C'.'City' AS 'City','A'.'Account_No' AS 'Account_No','A'.'Account_Type' AS 'Account_Type','A'.'Balance' AS 'Balance','A'.'Branch' AS 'Branch' from ('customer' 'C' join 'account' 'A' on (('C'.'Customer_ID' = 'A'.'Customer_ID')))) | utf8mb4 | utf8mb4_0900_ai_ci |
+-----+-----+
1 row in set (0.004 sec)
```

53. DESCRIBE VIEW

```
[mysql> DESCRIBE Customer_Account_View;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Customer_ID | int | NO | | NULL | |
| Customer_Name | varchar(100) | NO | | NULL | |
| City | varchar(50) | YES | | NULL | |
| Account_No | int | NO | | NULL | |
| Account_Type | varchar(20) | YES | | NULL | |
| Balance | decimal(12,2) | YES | | 0.00 | |
| Branch | varchar(50) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.007 sec)
```